

JOHN S. MCALISTER

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Education

- MAY 2026 **Ph.D. in Mathematics** — University of Tennessee - Knoxville
Mathematical Biology concentration
- MAY 2023 **M.S. in Mathematics** — University of Tennessee - Knoxville
Concurrent with PhD program
- MAY 2021 **B.S. in Zoology** — The Ohio State University
With Honors Research Distinction
- MAY 2021 **B.S. in Mathematics** — The Ohio State University
Math-Bio track

Research Interests

I study mathematical biology and game theoretic modeling. Using tools from PDEs, nonlocal equations and graph theory my main focus is the interaction between relational structure and game theoretic dynamics.

Papers

- In review **McAlister J.S.**, Burns N. (2026) *Myopic Best Response Replicates the Mean Curvature Flow*. Preprint <https://doi.org/10.48550/arXiv.2609.37803>
- In review **McAlister J.S.** (2026) *Structured Coordination on Planar Graphs: a Dual Approach*. Preprint <https://doi.org/10.48550/arXiv.2607.22865>
- In review **McAlister J.S.**, Fefferman, N.H. Mengesha, T.A. (2025) *Nonlinear Nonlocal Diffusion Equation for the Analysis of Continuous Coordination and Anti-coordination Type Games*. Preprint. <https://doi.org/10.48550/arXiv.2506.13929>
- October 2026 Dadlani R. **McAlister J.S.**, Schwarze A., Kawikatsu M., Iams S. Eissa T.L., Fefferman N.H. (2026) *Planned Behavior, Perceptual Biases, and the Dynamics of Collective Action*. Physical Review E <https://doi.org/10.48550/arXiv.2506.13929>
— Co-first author with R. Dadlani as a mentor for an undergraduate research project
- December 2025 **McAlister J. S.**, Brunner, J.L., Galvin, D. J., Fefferman, N.H. (2025) *A Game Theoretic Treatment of Contagion in Trade Networks*. PLOS computational Biology. <https://doi.org/10.1371/journal.pcbi.1013845>
- April 2025 Kirkland, S., Li, C., **McAlister, J.S.**, and Zhang, X. (2025) *Edge Addition and the Change in Kemeny's Constant*. Discrete Applied Mathematics. <https://doi.org/10.1016/j.dam.2025.04.031>
- January 2025 **McAlister, J.S.**, Fefferman N. H. (2025) *Insights into the coordination game with neutral options through simulation*. Dynamic Games and Applications <https://doi.org/10.1007/s13235-024-00612-4>
- January 2025 **McAlister, J.S.**, M.J. Blum, Y. Bromberg, N.H. Fefferman, Q. He, E. Lofgren, D.L. Miller, C. Schreiner, K. Selcuk Candan, H. Szabo-Rogers, and J. M. Reed (2025) *An Interdisciplinary Perspective of the Built-Environment Microbiome*. FEMS Microbiology Ecology <https://doi.org/10.1093/femsec/fiae166>

- December 2023 Fefferman, N.H., **McAlister, J.S.**, Akpa, B.S., Akwataghibe, K., Azad F.T., Barkley K., Bleichrodt, A., Blum M.J., Bourouiba, L., Bromberg, Y., Candan K.S., Chowell, G., Clancey, E., Cathroan, F.A., DeWitte, S.N., Fernandez, P., Finnoff, D., Flaherty, D.T., Gibson, N.L., Harris, N., He, Q., Lofgren, E.T., Miller, D.L., Moody, J., Muccio, K., Nunn, C.L., Papeş, M., Pachalidis, I.Ch., Pasquale, D.K., Reed, M.J., Rogers, M. B., Schreiner, C. L., Strand E.B., Swanson C.S., Szabo-Rodgers, H. L., and Ryan, S. J. (2023) *A New Paradigm for Pandemic Preparedness*. Current Epidemiological Reports. <https://doi.org/10.1007/s40471-023-00336-w>
- April 2022 **McAlister, J.S.**, Hamilton, I. (2022) *An Adaptive Dynamic Model for the Vigilance Game in Group Foragers*. Journal of Theoretical Biology. 538:111033. <https://doi.org/10.1016/j.jtbi.2022.111030>

Invited Talks

- April 2026 **McAlister J. S.** (2026) *Non-local models in Eco-Evolutionary Processes* Interfaceing Ecology and Evolution. Mini Symposium at Princeton University, Princeton, New Jersey
- February 2026 **McAlister J. S.** (2026) *On Structured Models of Coordinating Systems* Invited talk for the Arizona State University School of Math and Natural Sciences, Phoenix Arizona
- August 2025 **McAlister J. S.** (2026) *Replicator Dynamics with Spatial Structure* Invited talk for the NSF APPEX center summer Research Symposium, Knoxville Tennessee

Posters and Presentations

- February 2026 **McAlister J. S.** (2026) *Structured Coordination Games in the Plane* Talk given in the Geometric Analysis Seminar at the University of Tennessee
- January 2026 **McAlister. J. S.** (2026) *Replicator Dynamics for Games with Relational Structure* Talk given in the AMS Contributed Paper Session on Operations Research, Game Theory, and Social Sciences at The Joint Math Meetings 2026 in Washington DC
- December 2025 **McAlister. J. S.** (2025) *Replicator Dynamics for Games on Networks* Talk given at The Conference for Complex Networks and Applications 2025 at SUNY Binghamton
- September 2024 **McAlister. J. S.** (2024) *Structured Coordination in Continuous Spatial and Strategic Domains* Talk given at AMETHYST: Game Theory in Complex Systems during the Conference on Complex Systems 2024 at the University of Exeter
- March 2024 **McAlister J. S.** (2024) *The Structured Coordination Game with Neutral Options* Talk given at The Mathematical Association of America - South East Section Meeting at the University of Tennessee - Knoxville
- April 2023 **McAlister J. S.** (2023) *An Adaptive Dynamic Model for a Vigilance Game among Group Foragers*. Talk given at the SIAM Graduate Research Showcase at the University of Tennessee-Knoxville

Funding

Proposals

- April 2024 **New Techniques for the Analysis of Coordination in General Discrete and Continuous Domains** Submitted to DARPA DSO Critical Orientation of Mathematics to Produce Advancements in Science and Security (COMPASS) (DARPA-EA-25-02-03) **Selected to give Oral Proposal Package (OPP)**

Support

Spring 2025	GRA supported by NSF(DBI) #2312115 PIPP Phase II: Analysis and Prediction of Pandemic Expansion (APPEX)
Fall 2024	GRA supported by NSF(DEB) #2207922 Socioeconomic and Epidemiological Drivers of Pathogen Dynamics in Wildlife Trade Networks
Fall 2024	GRA supported by NSF(DRL) #2247074 Developing and Early Understanding of Contagion in Preschool- and Kindergarten-Aged Children
Fall 2023	GRA supported by NSF(CCF) #2200140 PIPP Phase I: Predicting Emergence in Multidisciplinary Pandemic Tipping-points(PREEMPT)

Awards

April 2026	Spike D. Tickle STEM Endowed Fellowship - Winner
—	Awarded for excellent research by a graduate student in a STEM field
February 2024	Eaves Teaching Award - nominee
—	Nominated for excellence in teaching among late career graduate students
April 2023	Eaves Teaching Award - Finalist
—	Awarded for excellence in teaching among early career graduate students
April 2023	Math GTA Teaching Excellence Fellowship - Winner
—	Nominated for commitment to further the teaching mission of the University.
August 2021	Academic Performance Assistantship - Winner
—	Awarded for meeting academic milestones in the PhD program early.

Workshops

August 2026	Evolutionary Games: Mathematical Theory and Biological Insights National Institute for Theory and Mathematics in Biology (NITMB): Chicago, IL.
May 2022	CBMS conference: Interface of Mathematical Biology and Linear Algebra University of Central Florida, Orlando, FL.

Mentorship and Teaching Experience

July 2024 - Present	Applied Math Undergraduate Research Mentor University of Tennessee - Knoxville, Knoxville, TN.
Aug 2022 - Dec. 2023	Graduate Teaching Associate - Instructor of Record MATH 113 University of Tennessee-Knoxville, Knoxville, TN.
Aug 2021 - May 2022	Graduate Teaching Assistant MATH 119, 125 University of Tennessee-Knoxville, Knoxville, TN.
Aug 2019- May 2021	Undergraduate Teaching Assistant MATH 1075, 1149, 1150 The Ohio State University, Columbus, OH.

Leadership and Volunteerism

2022-2024	Member —Graduate Teaching Assistantship Advisory Council
2023-2024	Senator —Graduate Student Senate
2022-2023	President —Math Graduate Student Council

Research Experience

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| September 2026 - Present | Postdoctoral Research Associate — Levin Lab
Princeton University, Princeton, NJ.
Advisor: Prof. Simon Levin |
| January 2022 - May 2026 | Graduate Research Assistant — Fefferman Lab
University of Tennessee - Knoxville, Knoxville, TN.
Advisor: Prof. Nina Fefferman |
| May 2018 - May 2021 | Undergraduate Researcher — Hamilton Lab
The Ohio State University, Columbus, OH.
Advisor: Prof. Ian Hamilton |
| May-Aug. 2019 | REU Fellow — Ecological Modeling of the Mississippi River Basin
University of Wisconsin - La Crosse, La Crosse WI.
Advisors: Prof. Robert Allen, Prof. Anita Baines, Prof. James Pierce, Prof. Greg Sandland |

Software Experience

Proficient

- R
- Python
- MatLab
- $\text{\LaTeX}$
- Mathematica

Experienced

- C#
- Java
- Maple
- JMP
- NetLogo